Monomial f(q) = a*qˆ9 Report

Polynomial

$$f(q) = aq^9$$

Naive

$$g_{\text{naive}}(X) = X^9 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 362880a\frac{\Delta}{\epsilon}^{8} q + 60480a\frac{\Delta}{\epsilon}^{6} q^3 + 3024a\frac{\Delta}{\epsilon}^{4} q^5 + 72a\frac{\Delta}{\epsilon}^{2} q^7 + aq^9$$

$$\begin{aligned}\operatorname{Var}(g_{\text{naive}}(X)) = {} & 6402373705728000a^2\frac{\Delta}{\epsilon}^{18} + 3201055170969600a^2\frac{\Delta}{\epsilon}^{16} q^2 \\ & + 266721677107200a^2\frac{\Delta}{\epsilon}^{14} q^4 + 8886333173760a^2\frac{\Delta}{\epsilon}^{12} q^6 \\ & + 158370992640a^2\frac{\Delta}{\epsilon}^{10} q^8 + 1745743104a^2\frac{\Delta}{\epsilon}^{8} q^{10} \\ & + 12809664a^2\frac{\Delta}{\epsilon}^{6} q^{12} + 62208a^2\frac{\Delta}{\epsilon}^{4} q^{14} + 162a^2\frac{\Delta}{\epsilon}^{2} q^{16}\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{naive}}) = {} & 6402373705728000a^2\frac{\Delta}{\epsilon}^{18} + 3201186852864000a^2\frac{\Delta}{\epsilon}^{16} q^2 \\ & + 266765571072000a^2\frac{\Delta}{\epsilon}^{14} q^4 + 8892185702400a^2\frac{\Delta}{\epsilon}^{12} q^6 \\ & + 158789030400a^2\frac{\Delta}{\epsilon}^{10} q^8 + 1763596800a^2\frac{\Delta}{\epsilon}^{8} q^{10} \\ & + 13245120a^2\frac{\Delta}{\epsilon}^{6} q^{12} + 67392a^2\frac{\Delta}{\epsilon}^{4} q^{14} + 162a^2\frac{\Delta}{\epsilon}^{2} q^{16}\end{aligned}$$

$$\operatorname{Bias}(g_{\text{naive}}) = 362880a\frac{\Delta}{\epsilon}^{8} q + 60480a\frac{\Delta}{\epsilon}^{6} q^3 + 3024a\frac{\Delta}{\epsilon}^{4} q^5 + 72a\frac{\Delta}{\epsilon}^{2} q^7$$

Unbiased

$$g_{\text{unb}}(X) = X^9 a - 72X^7 a\frac{\Delta}{\epsilon}^{2}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^9$$

$$
\begin{aligned}
\mathrm{Var}(g_{\mathrm{unb}}(X)) = {} & 3841424223436800a^2\frac{\Delta}{\epsilon}^{18} + 1920712111718400a^2\frac{\Delta}{\epsilon}^{16}q^2 \\
& + 160059342643200a^2\frac{\Delta}{\epsilon}^{14}q^4 + 5335311421440a^2\frac{\Delta}{\epsilon}^{12}q^6 \\
& + 95273418240a^2\frac{\Delta}{\epsilon}^{10}q^8 + 1058593536a^2\frac{\Delta}{\epsilon}^{8}q^{10} \\
& + 8019648a^2\frac{\Delta}{\epsilon}^{6}q^{12} + 44064a^2\frac{\Delta}{\epsilon}^{4}q^{14} + 162a^2\frac{\Delta}{\epsilon}^{2}q^{16}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{MSE}(g_{\mathrm{unb}}) = {} & 3841424223436800a^2\frac{\Delta}{\epsilon}^{18} + 1920712111718400a^2\frac{\Delta}{\epsilon}^{16}q^2 \\
& + 160059342643200a^2\frac{\Delta}{\epsilon}^{14}q^4 + 5335311421440a^2\frac{\Delta}{\epsilon}^{12}q^6 \\
& + 95273418240a^2\frac{\Delta}{\epsilon}^{10}q^8 + 1058593536a^2\frac{\Delta}{\epsilon}^{8}q^{10} \\
& + 8019648a^2\frac{\Delta}{\epsilon}^{6}q^{12} + 44064a^2\frac{\Delta}{\epsilon}^{4}q^{14} + 162a^2\frac{\Delta}{\epsilon}^{2}q^{16}
\end{aligned}
$$

Comparison

$$
\text{mean gap} = -362880a\frac{\Delta}{\epsilon}^{8}q - 60480a\frac{\Delta}{\epsilon}^{6}q^3 - 3024a\frac{\Delta}{\epsilon}^{4}q^5 - 72a\frac{\Delta}{\epsilon}^{2}q^7
$$

$$
\begin{aligned}
\text{variance gap} = {} & -2560949482291200a^2\frac{\Delta}{\epsilon}^{18} - 1280343059251200a^2\frac{\Delta}{\epsilon}^{16}q^2 \\
& - 106662334464000a^2\frac{\Delta}{\epsilon}^{14}q^4 - 3551021752320a^2\frac{\Delta}{\epsilon}^{12}q^6 \\
& - 63097574400a^2\frac{\Delta}{\epsilon}^{10}q^8 - 687149568a^2\frac{\Delta}{\epsilon}^{8}q^{10} \\
& - 4790016a^2\frac{\Delta}{\epsilon}^{6}q^{12} - 18144a^2\frac{\Delta}{\epsilon}^{4}q^{14}
\end{aligned}
$$

$$
\begin{aligned}
\text{MSE gap} = {} & -2560949482291200a^2\frac{\Delta}{\epsilon}^{18} - 1280474741145600a^2\frac{\Delta}{\epsilon}^{16}q^2 \\
& - 106706228428800a^2\frac{\Delta}{\epsilon}^{14}q^4 - 3556874280960a^2\frac{\Delta}{\epsilon}^{12}q^6 \\
& - 63515612160a^2\frac{\Delta}{\epsilon}^{10}q^8 - 705003264a^2\frac{\Delta}{\epsilon}^{8}q^{10} \\
& - 5225472a^2\frac{\Delta}{\epsilon}^{6}q^{12} - 23328a^2\frac{\Delta}{\epsilon}^{4}q^{14}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Bias}(g_{\mathrm{naive}})^2 = {} & 131681894400a^2\frac{\Delta}{\epsilon}^{16}q^2 + 43893964800a^2\frac{\Delta}{\epsilon}^{14}q^4 \\
& + 5852528640a^2\frac{\Delta}{\epsilon}^{12}q^6 + 418037760a^2\frac{\Delta}{\epsilon}^{10}q^8 \\
& + 17853696a^2\frac{\Delta}{\epsilon}^{8}q^{10} + 435456a^2\frac{\Delta}{\epsilon}^{6}q^{12} + 5184a^2\frac{\Delta}{\epsilon}^{4}q^{14}
\end{aligned}
$$